



Aim - To implement communication between client as

Ass - Java Applet - Animation of a Bouncing Ball.

Objectives

- To understand java Applet lifecycle method.
- To implement animation using graphics and Thread
- To simulate bouncing effect using logic.

Concepts used:-

- Applet (java.applet.Applet)
- Graphics (drawOval, fillOval)
- Multithreading (Thread)
- Animation using repaint().

Theory

A java Applet is a small program that runs inside a browser or applet viewer.

Animation is achieved by repeatedly updating object positions and repainting them.

In this exp, a ball moves inside the applet window. When it hits the boundary, its direction reverses, creating a bouncing effect.



This done using co-ordinate updates and conditional checks.

Algorithm:-

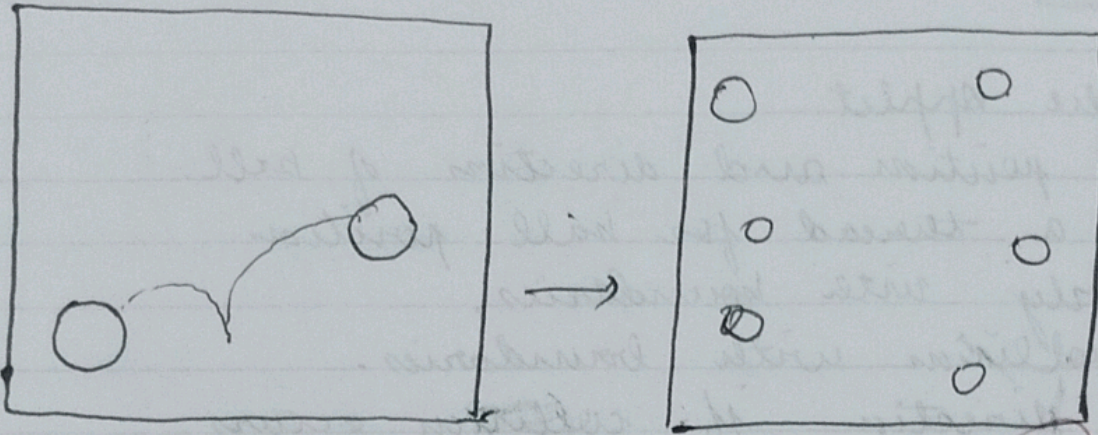
1. Start the Applet
2. Initialize position and direction of ball.
3. Create a thread for ball position.
4. Continuously with boundaries.
5. Check collision with boundaries.
6. Reverse direction if collision occurs
7. Repaint the screen
8. Repeat

Source Code:-

```
import java.applet.Applet;
import java.awt.Graphics;
```

```
public class BouncingBall
    implements Applet, Runnable {
    int x = 50, y = 50;
    int dx = 1, dy = 1;
    Thread t;
    public void init() {
        t = new Thread(this);
        t.start();
    }
}
```


output



Source Code:-

import java.awt.*;
import java.awt.event.*;

public class Applet extends Applet implements Runnable {

int x = 50, y = 50;

int dx = 1, dy = 1;

Thread t;

public void init() {

t = new Thread(this);

t.start();

}



```

public void run() {
    while (true) {
        x += dx;
        y += dy;
        if (x <= 0 || x >= getWidth() - 30)
            dx = -dx;
        if (y <= 0 || y >= getHeight() - 30)
            dy = -dy;
    }
}

```

```

    repaint();
    try {
        Thread.sleep(20);
    } catch (Exception e) {}
}

```

```

public void paint (Graphics g) {
    g.fillOval(x, y, 30, 30);
}
}

```

Result:-

The java Applet program successfully demonstrates animation of a bouncing ball.